

Ido Lavion

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1 Personal Information

- Name: Ido Lavion
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1.1 Languages

- Hebrew: Native
- English: Fluent
- Arabic: Basic (can read but not speak or listen)
- Japanese: Basic (Can speak and listen but not read or write)

2 Formal Education

2.1 University

- B.Sc. in Physics, predicted 2024-2027.
GPA: 93
In the 2nd year of my program, Studying in Israel's Open university.

2.2 Pre University

2.2.1 Hof Hasharon High School, 2024-2027

- GPA: 10th grade: 97, 11th grade: 93.5, 12th grade: -
10 units in multitech (Mechatronics, Mechanics and CAD), 5 Units in physics, 5 units in maths, 5 units in English, 5 Units in Computer Science

2.2.2 Academy In high School program, 2024-2026

- Accelerated Physics and Maths to finish them early via the program.

2.2.3 Bioinformatics and biotechnology program, 2024-2027

- GPA: 10th grade: 100, 11th grade: 97, 12th grade: -
5 Units in computational biology offered by the Davidson Institute of Science and Technology that is part of the Weizmann Institute of Science.

3 Work Experience

- Private tutoring, 2023-2024
Tutoring mostly elementary students in mathematics, hebrew, english and sciences.
- Summer job as a martial arts instructor, 2024
Teaching at a local summer camp for kids aged 6-12, focusing on self-defense and discipline.
Including organizing and making lessons, taking care of a group of 20-30 kids alongside 1 younger helper and overseeing the activities.

4 Works And Projects

4.1 Protein folding prediction using machine learning, 2026

- Developed a machine learning model to predict protein structures (of specifically beta-barrels) from amino acid sequences.
This project was done as my 11th grade final project in the bioinformatics and biotechnology program.

4.2 Prey and predator simulation, 2025

- Developed a simulation of prey and predator populations using PPO reinforcement learning.
This project was done as my 10th grade final project in the bioinformatics and biotechnology program.

4.3 TVC rocket, 2026

- Planned to develop and build a TVC (Thrust vector control) rocket that can be controlled in flight for low-altitude operations.
This project was supposed to be done as my 11th grade final project in the multitech program but was canceled due to the war in Israel.
There remains 3D models of the rocket parts and the rocket's physical design alongside the TVC control system design and relevant calculations.

4.4

5 Proficiencies and Interests

- Programming: Python, Java, Assembly (x86), C/C++ (Classically as well as for microcontrollers), MATLAB, Mathematica, TYPESCRIPT, HTML, CSS, SQL
- Software: OnShape, Blender, Unity, Visual Studio/Code, Arduino IDE, Adobe Creative Suite, Microsoft Office Suite, LaTeX
- Interests: Physics, Mathematics, Astronomy and Space exploration, Reading, Martial arts, Music, Video games, Programming and software development, Robotics, 3D modeling and printing.